Python Programming Fundamentals

Lists of Dictionaries

Programming Fundamentals Team

SETU

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Outline

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1. A List of Records

```python
movies = [
    {"title": "Jaws", "year": 1975, "rating": 5},
    {"title": "Alien", "year": 1979, "rating": 5},
    {"title": "Shrek", "year": 2001, "rating": 4}
]
```

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movies = [
    {"title": "Jaws", "year": 1975, "rating": 5},
    {"title": "Alien", "year": 1979, "rating": 5},
    {"title": "Shrek", "year": 2001, "rating": 4}
]
```

Each dictionary is one record.

The list stores the collection.

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2. Displaying Records

```
for movie in movies:
    print(movie["title"], movie["year"])
```

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for movie in movies:
    print(movie["title"], movie["year"])
```

Formatted output:

```python
for movie in movies:
    print(f"{movie['title']} ({movie['year']})")
```

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3. Searching

```python
search_title = input("Enter title: ")

for movie in movies:
    if movie["title"].lower() == search_title.lower():
        print("Found:", movie)
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Important idea:

- loop through each record
- check one field
- display the matching record

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4. Filtering

```python
for movie in movies:
    if movie["rating"] >= 5:
        print(movie["title"])
```

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```python
for movie in movies:
    if movie["rating"] >= 5:
        print(movie["title"])
```

Filtering means selecting only the records that match a condition.

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5. Adding New Records

```python
new_movie = {
    "title": input("Title: "),
    "year": int(input("Year: ")),
    "rating": int(input("Rating: "))
}

movies.append(new_movie)
```

5. Adding New Records

```python
new_movie = {
    "title": input("Title: "),
    "year": int(input("Year: ")),
    "rating": int(input("Rating: "))
}

movies.append(new_movie)
```

This is the Create step in CRUD.

Thanks for Watching - Any questions?